



“The bus is accessible, but how do you get to the bus”: First and last mile experiences of disabled transit riders

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ABSTRACT

To use public transit, riders must complete first and last mile trip segments. However, transportation planning measures of access to opportunity often assume that transit riders can complete first and last mile trips with ease. This paper contributes to the understanding of how disabled transit riders experience the first and last mile of public transit trips. Using a mobility justice framework, interviews with disabled transit riders reveal accessibility challenges along the first and last mile in Austin, TX and Seattle, WA. Participants in both Austin and Seattle faced myriad accessibility challenges along the first and last mile, indicating that transit network size and local politics did not influence travel experiences. Findings indicate disproportionate and intersectional barriers to accessing public transit in both cities, especially among female transit riders. I find that gender, social conditions, built environment quality, connectivity, and public engagement experiences influence access to transit for disabled people along the first and last mile. This work reveals how planners can better engage with disabled transit riders about their experiences and incorporate mobility justice goals to improve first and last mile accessibility.

1. Introduction

Public transit systems inherently prioritizes the travel needs of non-disabled commuters and makes assumptions about how transit riders complete first and last mile trip segments (Lubitow et al., 2017). To use public transit, riders must be able to reach stops and then find their way from stops to a destination in a safe and efficient manner. Assumptions about average users and their travel behavior in common transportation planning measures leave out many transportation system users, including disabled people, by ignoring how the built environment, existing land uses, social contexts, and pedestrian infrastructure influence travel behavior (Church and Marston, 2003; Omura et al., 2020). Not only do transit vehicles, stops, and stations need to consider accessibility for disabled people, but also the first and last mile pathways to and from transit need to meet the needs of potential riders (Liu et al., 2023).

Prior work in transportation planning has oversimplified how disability interacts with access and mobility, and a deeper understanding of the complexities of accessibility along the first and last mile is a needed step towards justice. *Accessibility*, in this case, refers to accommodations in design and function to meet the needs of disabled people. Approximately 13% of Americans identify as disabled (U.S. Census

Bureau, 2023). In addition, disabled people use public transit to commute to work more so than non-disabled individuals (4.3% and 2.7%, respectively) (Brumbaugh, 2018). Disabled people experience inaccessibility in a variety of ways, reducing their opportunity to be included in political, economic, and social activities (Bezyak et al., 2020; Clarke, P. J., J. A. Ailshire, E. R. Nieuwenhuijsen, and M. W. de Kleijn – de Vrankrijker, 2011). Missing sidewalks and inaccessible pedestrian signals at crosswalks are examples of transportation decisions that contribute to travel barriers for disabled people.

I use the phrase *access* to refer to the ease of reaching people or places, instead of the phrase *accessibility* to respect disability studies scholarship. Particularly when measuring equity, or the fair distribution of services, disability is typically addressed with a one-off measure of access to opportunities. The measure of whether a transportation system is usable for disabled people is often reduced to the number of accessible transit stops or vehicles (Grise et al., 2018; Zhang et al., 2023). This approach misses the first and last mile trip segments as well as other barriers to travel, like unwelcoming social climates. Additionally, there are several challenges at the social and institutional levels that hinder access to transit for disabled people compared to other marginalized groups. Social exclusion, data gaps, compliance-based approaches, and a limited conceptualization of disability in transportation planning have

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perpetuated travel disadvantages (Levine and Karner, 2023).

Including the accessibility of the first and last mile when evaluating access portrays the disability community's needs and values more accurately, leading to inclusive and equitable streetscapes (Wagner et al., 2023). Advancing mobility justice in practice can only be achieved if affected communities co-lead the planning process. Without meaningful public engagement from the communities who have been historically and are currently harmed by planners, transportation decision-makers will perpetuate inequitable outcomes. Therefore, speaking directly with disabled travelers who rely on public transit for their travel needs is imperative to better understand experiences along the first and last mile.

The diversity of disability and lack of data about accessibility presents challenges for planners, however, we cannot continue to simplify the needs of disabled people. A needed step towards equitable change for the disability community is to understand how current transportation system conditions and accessibility affect mobility and access. This paper contributes to the limited body of research on the travel experiences of disabled pedestrians by answering the following research question: how do social and built environments influence access to transit along the first and last mile for disabled people? To evaluate first and last mile travel experiences, I conducted 16 interviews with disabled transit riders located in Austin, TX, and Seattle, WA.

This work uses a mobility justice framework to acknowledge the overarching social, political, and physical structures that restrict individual autonomy. Mobility justice considers how individuals of all abilities achieve access and mobility and considers how the built environment reproduces privilege and power (Sheller, 2020; The Untokening). This multi-scalar approach to justice considers both personal limitations to mobility, such as accessibility or the risk of being hit by a vehicle, as well as the broader context of neighborhoods and urban regions. In the broader sense, prioritizing certain road users over others, especially car users, creates injustices during the planning process, in resultant travel behavior, and in the design of our urban forms.

I find that intersectionality, social context, built environment quality, confidence, and relationships with public agencies influence access to transit for disabled people along the first and last mile. Despite differences in context, participants in both cities had similar accessibility challenges and experiences along the first and last mile. These findings indicate that planners can learn how to improve accessibility along the entire trip chain by engaging with disabled transit riders about their experiences and by using mobility justice to prioritize inclusivity in first and last mile planning practices.

2. Literature review

The following literature review is structured to summarize previous work related to the first and last mile and the travel behavior of disabled people. I first discuss the historical and current challenges for public transit planning, including the inequitable distribution and quality of service. Then, I link transportation research with disability to highlight travel experiences and barriers. This review reveals the need to be more inclusive of disabled people in the entirety of the transportation planning process and provides motivation for this research.

2.1. Public transit planning

Accessibility, affordability, and privilege, and choice are central to understanding how public investments, like public transit, are related to mobility justice (Stafford, 2020; Bullard and Johnson, 1997). Public transit systems connect people to opportunities using space-efficient and relatively affordable modes, providing low-cost mobility for those who cannot or do not want to drive. The decision for a potential transit rider to use the system, however, depends on their perception of the local environment and their capacity to actually reach a transit stop (Liu et al., 2018). Features of the built environment and urban form, including

street network design, intersection density, proximity to destinations, parking availability, and land use diversity, are extremely relevant to public transit trips (Ewing and Cervero, 2010; Chatman, 2015). The quality and condition of built environment features are amplified for disabled people, and urban residents report more severe disabilities when the urban built environment is unfavorable (Clarke et al., 2008; Clarke and George, 2005; Rosso et al., 2011).

In addition, the presence of other people and social interactions affect travel behavior. The social nature of public space results in perceptions, attitudes, and acceptance that influence how individuals feel while traveling. These different barriers coupled with social identities and oppressions play a significant role in travel behavior, particularly regarding public transit use (Iudici et al., 2017; Lubitow et al., 2017). Disabled people face insecurity, exclusion, and ultimately travel less because of the social conditions they experience in public spaces (Iudici et al., 2017).

Public transit investments are also tied to gentrification and displacement. Active transportation infrastructure improvements occur in communities with less visible minorities, fewer children, and closer to the urban core (Fischer and Winters, 2021). While improving transit service benefits riders, neighborhoods can be subject to unprecedented new demands that increase costs and change neighborhood structures, pushing out low-income residents and other marginalized groups (Sandoval, 2018). Transportation and land use policy actions, such as zoning, subsidies, and economic structures, also contribute to social inequities and travel outcomes (Zuk et al., 2018).

2.2. First and last mile environments

The discussion of the first and last mile trip segments is growing in transportation planning literature in light of agency goals related to sustainability and equity. Shifting travel behavior from personal vehicle dependency to public transportation and other modes involves a deep dive into how people traverse the first and last mile. Multiple modes of transportation are often used to travel from origins to destinations, such as walking to transit, riding transit, and then walking to a final destination. Ruscher et al. (2018) argues in favor of accessible intermodal transportation, including the integration of on-demand services in complement with traditional public transportation. Similarly, Boarnet et al. (2017) examine the relationship between job access and first and last mile accessibility and reveal the importance of this segment of the trip chain compared to other factors (i.e. frequency or headway). Therefore, public transportation agencies must work cooperatively with other entities to ensure that potential riders can access their system along the first and last mile.

Public transportation agencies commonly address first and last mile challenges through targeted investment with local agencies to improve pedestrian infrastructure or through public-private partnerships. Agencies also utilize private transportation network companies, like Uber and Lyft, to provide first and last mile connections between destinations (Liu et al., 2018). While Uber and Lyft offer wheelchair-accessible vehicles, they continue to fall into a gray area under the ADA and are not required to offer this service in all markets (Meshel, 2021). Other regions have replaced fixed-route transit with microtransit, which is an on-demand, dynamically-routed rideshare service usually offered in partnership with transportation network companies (Brumfield, 2023). By offering all service in a door-to-door model, this eliminates the need for first and last mile improvements. Micromobility, another emerging transportation option, tries to address the first and last mile trip segments by providing alternative options to walking. Micromobility can include e-scooters, shared bicycles, and shared shuttles. However, due to the recent implementation into urban environments, questions regarding regulation, design, purpose, parking, storage, safety, and use have yet to be answered (Tice, 2019). The direct exclusion of unbanked populations, individuals without smartphones, and disabled people has not been sufficiently researched and needs

attention.

2.3. Disability and travel

Disabled people experience inaccessibility in a variety of ways, reducing their opportunity to be included in political, economic, and social activities. The Americans with Disabilities Act (ADA) of 1990 was a significant step towards accessibility to public facilities, however, barriers in the built and social environments continue to affect mobility for disabled people. Even though public agencies build new infrastructure under ADA guidelines, these minimum standards are not adequate for barrier-free travel (Levine and Karner, 2023). The lived experience of disability is largely missing from transportation planning practices that are based on meeting minimum standards for accessibility set by the ADA. The ADA’s implementation at the local scale is largely limited to political and institutional applications of what the government deems acceptable accommodations for disabled people, rather than stemming from disability activism and lived experience (Hamraie, 2018).

Disability does not exist in isolation, but is instead an experience constructed by the built and social environments (Jackson, 2018). The limitations and exclusions disabled people face when interacting with transportation systems and other people in public spaces construct their disability. For example, a public transit stop with only visual signage about the bus route is inaccessible to a visually impaired person. Tactile and auditory features to communicate in alternative ways would make that transit stop accessible to more people. From an urban environment designed for non-disabled people to discrimination from vehicle operators, simply getting to and boarding a transit vehicle can be a major challenge for this group (Lubitow et al., 2017). There are also health and safety considerations for disabled pedestrians. Disabled pedestrians are at higher risk of injury and fatality than other communities because of slower walking speeds, difficult crossing intersections, and environmental factors (Schwartz et al., 2022).

Despite these well-known issues, the literature has only scarcely engaged with the transit user experience for disabled people (Aghaabasi et al., 2018; Sharifi et al., 2016; Stafford and Baldwin, 2018). In addition, existing scholarship focuses on barriers to physical activity for disabled adults, largely adults with mobility disabilities. Kirchner et al. (2008) administered surveys to understand environmental and social barriers to physical activity and found that more planning and strategizing is required to walk or roll in their community. Rosenberg et al. (2013) examined how the built environment influences the physical activity of adults with mobility disabilities and found that built environment quality was a significant barrier to participation. However, there is a gap in transportation planning scholarship regarding the impacts of disability and disability type when accessing public transit stops and stations. Understanding how disability type relates to travel behavior is important because transportation research typically focuses on wheelchair users with a mobility disability. However, impairments to vision, hearing, learning, and self-care can impact walking, driving, wayfinding, navigating, and decision-making. Transit riders with multiple disabilities, chronic conditions, vision impairments, and cognitive impairments have reported more accessibility challenges when using public transit than riders with mobility disabilities (Bezyak et al., 2020).

3. Methods and data

I conducted a comparative case study to understand access to transit for disabled people along the first and last mile in Seattle, WA and Austin, TX. Seattle is an exceptional case due to its greater than average investment in sidewalk condition data collection, widespread public transit infrastructure, and city-wide focus on accessibility. In 2017, Seattle conducted a detailed sidewalk assessment to track barriers to accessibility across all public sidewalks (Seattle Department of Transportation, 2018). This investment highlights the region’s focus on improving conditions for disabled travelers. Additionally, the mode

share for commuting by public transit in Seattle is at 16% (U.S. Census Bureau, 2023). Austin is a typical case and generalizable to the other mid-to-large-sized cities in the United States, with limited sidewalk data collection and centralized transit infrastructure investments, especially along the north-south corridor of the city. Only 2.6% of workers commute using public transit in Austin (U.S. Census Bureau, 2023). I capture first-person accounts of disabled people’s experiences along the first and last mile in both cities, considering how physical, social, and environmental structures that can either facilitate or impede travel.

This case study analyzes primary data from semi-structured interviews with disabled transit riders. Qualitative methods are central to understanding the lived experiences of marginalized communities, including disabled people. Disability fluctuates over time and reflects diverse layers of intersectional discrimination. Research about the travel experiences of disabled people must come directly from meaningful engagement opportunities. Rather than using additive analyses of different social identities, oppressions, or privileges, where groups are ranked or scored, we can acknowledge how social systems interact and coexist through inclusive and participatory research designs (Berger and Lorenz, 2015; Clifton and Handy, 2003; Windsong, 2018). Qualitative research practices can employ an intersectional lens, helping to make meaning from the intricate lived experiences of disabled people.

I led semi-structured interviews focused on first and last mile accessibility with 16 disabled individuals who rely on public transit for their transportation needs between September and December 2022. Interviews were conducted virtually using Zoom and lasted 30–60 minutes on average. Interviews were recorded, transcribed, and coded to synthesize data and extract themes. Interview participants were over 18 years old, lived in the greater Austin, TX or Seattle, WA region, and self-identified as a disabled transit rider. My sampling approach was not to recruit people to cover every possible disability type, but rather to include the voices of diverse individuals who have been historically excluded from transportation planning research. I used purposive and snowball sampling to recruit participants with varied identities, backgrounds, and types of disability. I contacted professional networks, community-based organizations, civil rights organizations, university student organizations, community centers, and public transit agency advisory groups to recruit interview participants by email. Initial email contact included a link to schedule a virtual interview and the option to connect me with other individuals who may also be interested in the interview process. All participants were able to select their interview date and time using the scheduling tool *Calendly*.

Table 1 describes the characteristics of interview participants. My sample included more participants located in Austin, and slightly more females than males. Most participants identified as White, followed by Black, Latinx, and Asian. Mobility and vision disabilities were most common in my sample, followed by multiple disabilities and cognitive

Table 1
Transit rider interview characteristics.

	N = 16
Location	
Seattle	6
Austin	10
Gender	
Male	7
Female	9
Race	
White	10
Black	3
Latinx	2
Asian	1
Disability Type	
Vision disability	5
Mobility disability	6
Cognitive disability	1
Multiple disabilities	4

disabilities. My interview sample included individuals recruited through the organizations listed in Table 2. I built relationships with each organization listed in Table 2 to connect with disabled transit riders in each city. I acknowledge that my interview sample may be more educated and aware of first and last mile accessibility challenges because of their involvement with the organizations listed.

Each interview began with an introduction and summary of the research project. After I reiterated the key points of the informed consent document, recording began. I used an approved semi-structured interview protocol to organize and guide interview discussions. Interviews began with participants introducing themselves, their disability, and their typical travel behavior. The approach to interviewing did not classify participants beforehand, but instead, allowed for individuals to express their own relationship to disability and travel experiences. Additionally, interview conversation was not structured around mobility disability, but rather, how all disability types may impact mobility and accessibility. By having interviewees self-identify and describe their travel experiences along the first and last mile, I am challenging established norms, and categories in transportation planning scholarship that typically focus on mobility disability. Questions centered around experiences along the first and last mile, accessibility of public spaces and the built environment and experiences using public transit.

I manually transcribed and inductively coded the interview data to perform a thematic analysis. A codebook was developed to establish reliability and consistency in data analysis. I used descriptive codes, emerging from interview context and topics, as well as *a priori* codes from prior knowledge about first and last mile accessibility. The codebook was refined to identify patterns and connections among interview data. Using an iterative process, codes were summarized into eight categories and five themes. Coding was validated through a peer-review process with colleagues versed in qualitative research methods. The peer-reviewers were selected based on their expertise in transportation planning and policy, social justice, and critical disability studies. Fig. 1 illustrates how the data analysis process of qualitative coding leads to meaning making and emerging insights. Codes are grouped into related categories, which nest into respective themes.

Interview limitations are important to acknowledge. The disabled transit riders recruited for this dissertation research are likely to be more aware and knowledgeable about first and last mile accessibility issues compared to the general public. I used disability and transit advocacy organizations and networks to recruit participants, indicating that they were already involved in local first and last mile issues. In addition, participants were recruited by email, and disabled transit riders without internet access are missing from this research.

4. Results

4.1. Intersectionality influences travel for disabled people

The experiences of respondents in this study suggest that individuals who identify as a person of color experience greater accessibility and access to opportunity challenges than other groups, an observation that is consistent with prior literature. Affordable housing choices and land

use decisions influence the travel behavior of disabled people. Socio-economic disadvantage is more prevalent among disabled people in the U.S. because of lower employment rates, fewer transportation options, and social exclusion (Hine and Grieco, 2003; Steinfeld et al., 2018). Two externalities related to affordable housing, residential location choice, and the first and last mile were prevalent in interviews. First, the decision to live near accessible infrastructure and public transit is often not financially possible, resulting in decreased travel overall and less opportunity. A participant with vision disabilities said:

Seattle is so interesting because it is so expensive...basically, the entire city...is unaffordable...I think that really impacts where people with disabilities can live. Because if you're moving into the city now, there's basically nowhere.

Secondly, the impact of living far from the urban core resulted in worse first and last mile experiences for interviewees due to inaccessible infrastructure. Another vision impaired participant explained:

So if you're in a rural part and get off on a bus at that last mile, it's a challenge because that's on the bottom of our list of making things ADA accessible for wheelchairs.

Three interviewees who identified as disabled women with mobility, vision, and cognitive disabilities described different safety concerns along the first and last mile. Regarding safety and the built environment, one participant revealed that the urban core's accessible infrastructure made her feel safer while using active and public transportation modes. As she traveled beyond downtown Austin, she experienced greater accessibility issues and felt more uncomfortable while traveling.

In terms of social safety, another female participant in Austin described her experiences with crowded pedestrian spaces, public transit stops, and public transit vehicles. Since the pandemic, she revealed that Capital Metro has reduced bus frequencies, resulting in larger crowds along the first and last mile and while riding transit. For her cognitive disability, a more calming environment would improve accessibility, stress, and her overall travel experience. She said, "There's also...no space... dedicated to sensory friendly spaces...I think those would be really nice to have."

Female interview participants also described stressful social phenomena that they experienced along the first and last mile. Two interviewees in Austin felt excluded due to interactions with other individuals along the first and last mile of their trips. A participant with a mobility disability described an experience with using her bicycle as her first mile transportation mode, before riding public transit. When she arrived at the bus stop, the bicycle rack attached to the front of the vehicle was already full. Her disability limits her ability to lift the heavy bicycle onto the passenger area of the bus, so her only option was to increase her travel time and wait for the next bus. She considered asking the operator to lower the bus as they do for wheelchair users, but she didn't feel confident in doing so. She said:

I...don't feel comfortable [lifting my bike] because...with my physical situation and the type of bike I have...I bet the bus driver could lower the bus like they do...trying to treat me like a wheelchair customer...but...I would feel too uncomfortable to ask.

The same participant described another incident where the transit operator made her feel uncomfortable when she took additional time to place her bicycle on the bus rack. She explained:

Table 2
Organizations contacted to recruit participants.

Seattle
Disability Rights Washington
Sound Transit – Citizen Accessibility Advisory Committee
The ARC of King County
Austin
Capital Metro – Access Advisory Committee
City of Austin – Bicycle Advisory Council
Longhorn TIES (Transition, Inclusion, Empower, Success)
National Federation of the Blind Texas

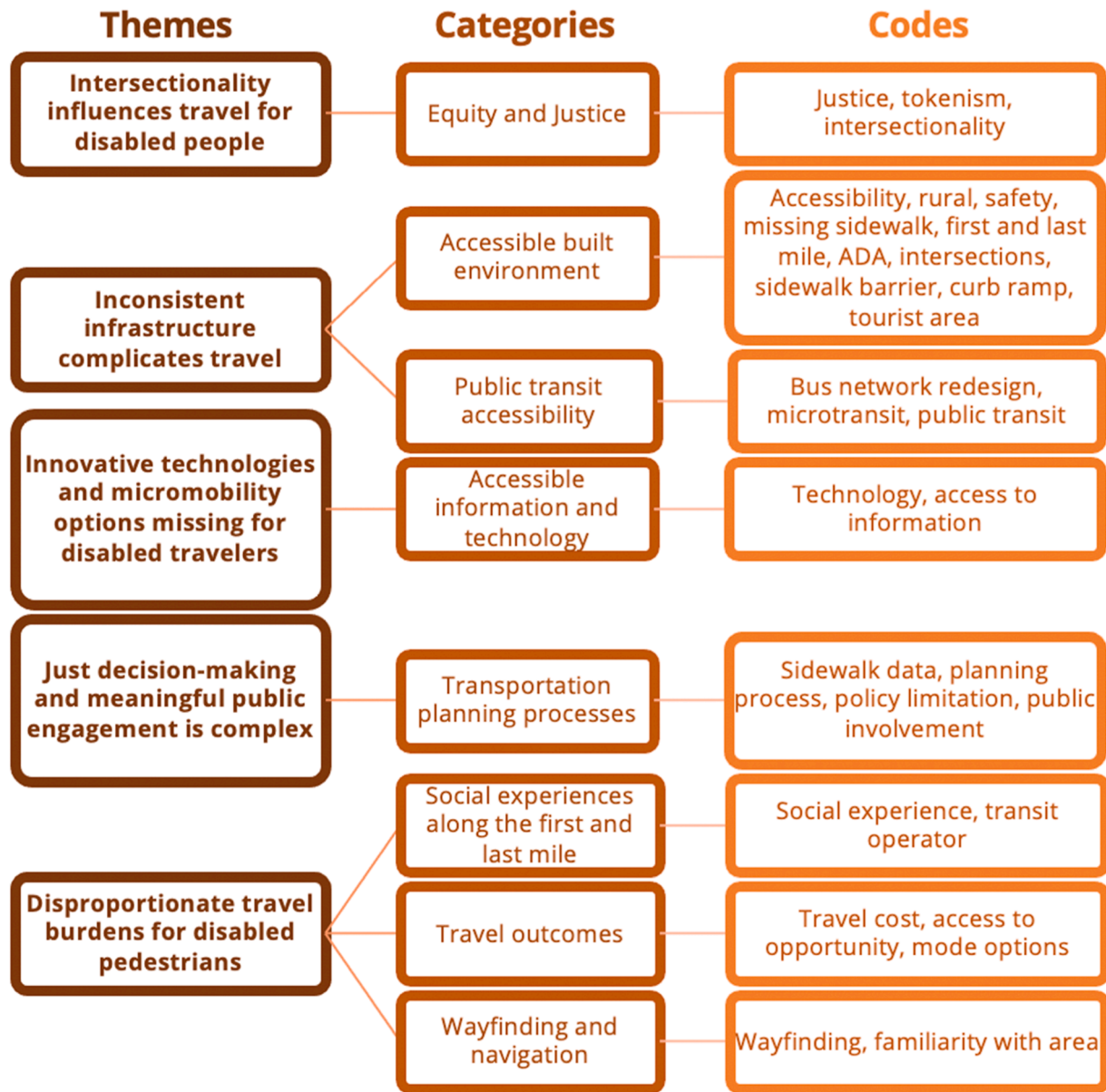


Fig. 1. : Hierarchy of codes, categories, and themes from interview data.

It's like kind of hard to put your bike on the bus rack... I'm not super strong. My bike is pretty heavy, and like I have one hand, and so I'm trying to navigate this, and the bus driver was just like really annoyed...it just made me feel really horrible. And honestly, it would have been enough to make someone else just never do that again.

Another female participant with a cognitive disability discussed her feelings of aloneness, saying that her peers are unaware of her experiences and uninformed about invisible disabilities. She said:

Invisible disabilities...are not usually not something that are considered a lot and students around us don't really know that we like have disabilities...and they are not informed.

4.2. Inconsistent infrastructure complicates travel

All interview participants, regardless of location, mentioned that accessible first and last mile infrastructure is inconsistent across their region. One participant summarized their experiences regarding first and last mile as inconsistent, explaining how local agencies focus on making vehicles and buildings accessible but forget about the travel

pathway to reach the facility. They said, "The bus is accessible, but how do you get to the bus?"

Pedestrian infrastructure presence and quality were another key factor for interviewees regarding their first and last mile experiences. An interviewee with a mobility disability described their experience with inadequate sidewalk quality, that coupled with their mobility disability, makes crossing the street to their nearest light rail stop difficult. She said:

I live within a mile of a...light rail stop. And it's not a long distance from the bus stop to the light rail station. But my challenge is actually crossing the street to get to that light rail station...I don't get across in allotted time because the crosswalk itself...it's not maintained.

A participant with a mobility disability discussed a related experience about the poor quality of sidewalks in their neighborhood. They stated:

[The] sidewalk it's not even ambulatory, let alone wheelchair friendly. There's all kinds of cracks in it, so I could slip and fall in it. And I don't think I could push myself that far by myself, [in] my chair.

Some participants indicated changing their travel path, sometimes in

lieu of safety, to avoid missing or poor-quality sidewalks. An interviewee with a vision impairment said:

Because I utilize a cane...I'm very good at walking on the side of the road real close, staying close to the curb...but I'd rather do that because it's smoother than walking in the grass.

The placement and accessibility of public transit infrastructure is also not consistently located along the sidewalk, adding additional challenges for vision-impaired travelers. The placement of bus stops either on the near or far side of an intersection, the location of bus stop signage and information, and the location of elevators, curb ramps, and crosswalks were identified as inconsistent by several vision impaired participants. They indicated that having consistent placements of these facilities would improve wayfinding and navigation along the first and last mile.

4.3. Innovative technologies and micromobility options missing for disabled travelers

Technology and real-time information have helped improve the accessibility of information for disabled people. All blind and limited-vision participants mentioned how fixed-route transit information along the first and last mile is available in multiple formats, including auditory announcements as vehicles are approaching and interactive displays at stops with different communication formats. However, one participant in Seattle explained that in high traffic areas, the audio communication was impossible to hear. They said, "When you got seven lanes of cars going back and forth...and you go tap your card on the machine as a blind person, you can't hear the tone."

New transit features and online presence was mentioned by a man with visual impairments in Seattle. He said that Seattle's new light rail system, named LINK, creates confusion when he uses a screen reader on the transit agency's website. The name of the light rail system is read by the screen reader in the same fashion as a clickable hyperlink, so navigating the desktop site or mobile smartphone application is challenging. This slows down his trip planning and could have been avoided if the agency chose a different name for the service. However, when he raised his concerns, he felt devalued by the transit agency. He said:

I tried to get them to change that because being a blind person using a screen reader on a web page, every time it says link, you think you can click on it. They didn't care. They were set on LINK.

Using micromobility options to traverse the first and last mile was another common thread of discussion among interviewees. Several participants said that they use taxis and transportation network companies on occasion to complete the first or last mile. Participants in both Austin and Seattle talked about microtransit pilots that had happened or are ongoing in their region, offered by local transit agencies. In Austin, a wheelchair user living in northwest Austin would welcome the Pickup by Capital Metro service to her neighborhood because it would expand her transportation options. She said, "I'm hoping they could come to my area of my neighborhood, and maybe I could use Pickup too."

Others indicated that they wish they had a wider variety of accessible micromobility options. Regarding e-scooters on campus as a form of micromobility, a participant with multiple disabilities said:

There's...language that just talks about...college students being lazy for using scooters for last mile accessibility...but people actually need that...like accessibility wise, there needs to be options available for that last part of the trip.

4.4. Just decision-making and meaningful public engagement is complex

While first and last mile experiences are largely dictated by social and built environment factors, equitable decision-making and just processes influence travel as well. A participant in Seattle complained about the accessibility conditions of the city's office building, saying that the location was inequitable for disabled people to visit. They said:

The city of Seattle's building...that's really an inaccessible location. The

only reason why they're there...is because the city owns the tower, which means the rent is free...They really don't have an incentive to move to a more equitable location, although they could.

An interviewee in Austin mentioned the complexity of improving accessible pedestrian spaces alongside other transportation planning priorities. Infrastructure decision-making is multifaceted due to the needs of different modes, travelers, and regions. Determining which investments are fair and just is not black and white. They said:

Anytime that people push for things that are more pedestrian friendly...they might inconvenience drivers...so people are always there to remind you that working class people need to drive...and that even includes some disabled people as well. There are other people who can drive but they can't walk that far...that is potentially a challenge for disability justice.

Another participant had a similar sentiment about a lack of accessible pedestrian infrastructure, saying:

I think it's a fundamental issue that stems from a lot of the city essentially being designed with cars first and pedestrians...have often been an afterthought at most...and then changing that is complicated.

The experiences of interviewees in this study suggest that disabled travelers are seldom in the room when transportation decisions are made and that they perceive accessibility as an afterthought for agencies, less important than other priorities. An interview participant in Austin expressed that the public engagement process for active transportation planning is largely led by white, male stakeholders. They said, "Something that was frustrating the whole time was how white and male-dominated [the process was]." The same participant mentioned that the individuals with strong voices in active transportation planning were removed from the issues they spoke about. They said:

I feel like people who don't rely on transit are coming up with these plans. You know they're just talking about the commute that's already outdated.

Another interviewee recalled a recent transit pilot project's public engagement process. They were frustrated by the lack of communication after the pilot ended, and said:

They put all this effort into having community spaces where community can give their input, give their feedback, or...help co-create...They're willing to entertain a pilot. But when the pilot is over, you hear nothing...I feel like transit agencies...They have all this energy in the beginning, in the middle to work...and communicate with community. But then when it comes to the end, it's crickets.

Meaningful public engagement, where marginalized communities are given power to set priorities and have a voice in decision-making, significantly differ from agency-led engagement practices (Linovski and Baker, 2023). Representation of populations served is not enough to justify equitable participation; rather, acknowledging historic and ongoing power disparities between groups in equity discussions and using this frame to make decisions can advance justice (Linovski and Baker, 2023).

Diversifying the transportation planning profession was also mentioned by a participant in Seattle. They explained that less privileged individuals who are unable to obtain a graduate degree or unpaid internship are bound from working in the field, helping to make change. They believed that education barriers, related to job qualifications, contributed to the lack of inclusivity in transportation planning. They said:

The pathways that exist for professional opportunity...are really relegated to students...so if you want to work for the city, you basically have to have a degree in engineering, urban planning, social work, or policy...there's really no equitable pathway to employment.

The same individual discussed issues with race in the transportation planning field and believed that this was the only form of diversity considered. Diversity in education, experience, and ability, are missing from the field. They said:

In terms of having more inclusivity within transit, you have to talk about diversifying your workforce beyond race, which I know is really hard for people...We have a very narrow way of thinking about who's qualified. And

to me, that's a leadership problem.

Lastly, the inability to drive was mentioned as a barrier to entry for public service professions. The interviewee considers the city's driver's license requirement for employment to be discriminatory against both disabled people and nondrivers. They stated:

Nearly all of their positions...required a driver's license, even for positions that didn't require driving...[but] there's a lot of people who don't have driver's licenses that you're basically excluding from applying...you're discriminating against two classes of people for no reason.

4.5. Disproportionate travel burdens for disabled transit riders

All interviewees mentioned that disabled travelers experience disproportionate travel burdens along the first and last mile. Several participants revealed that these burdens actually change their travel behavior, in terms of travel time or travel cost. A participant with a mobility disability in Seattle stated:

The sidewalk infrastructure that leads from the transit station to the municipal tower is awful...So what do I do? I go one extra stop south out of my way and I go up a different street to get to that building. But it's a lot of extra energy and it's about six blocks out of my way for nothing.

Another interviewee described the inconvenience of altering travel behavior to avoid missing sidewalks along the first and last mile. Having to change travel plans due to inaccessible first and last mile conditions is an inconvenience specific to the disability community that can be mitigated through improved planning. They said:

That's really an inconvenience for an individual with a disability, whether it's time or distance... that's very tedious and cumbersome.

A wheelchair user explained how adaptable their travel behavior needs to be due to accessibility concerns. They stated:

As a wheelchair user, I have to juggle inaccessibility frequently. This could be due to construction, a delivery driver parking on a curb ramp, one of those scooters that people lay on the sidewalk, or another reason...people don't really understand the consequences of blocking a sidewalk or curb ramp. There has been at least one time where I had to take the bus one stop just to get around the obstacle.

5. Discussion

This work also contributes to the expanding literature on intersectionality, gender, and travel behavior. Female interview participants experienced additional challenges, and perceived accessibility challenges along the first and last mile to be a greater burden than for other participants. Inequalities for disabled people, including overall health and well-being, are largely driven by social factors and experiences (Aitken et al., 2022). Women share disparate travel experiences, and disability compounds this phenomenon (Judici et al., 2017). Safety, both physically from other street users like drivers and socially from other transit riders and operators significantly impacted their travel experiences. This is consistent with academic literature about gendered travel experiences and public transit (Chowdhury et al., 2023; Judici et al., 2017; Rosenbloom, 2021). However, this finding differs from the previous work on physical activity for older, disabled adults that indicated most study participants felt safe while walking in their neighborhoods (Rosenberg et al., 2013). This difference may be due to the focus on public transit access, rather than opportunities for physical activity more broadly. The specific trip purpose of walking/rolling to reach a public transit stop may present different safety and social concerns than walking/rolling for health or leisure. The familiarity of a residential neighborhood may also feel safer for a pedestrian compared to an urban region with mixed land uses and more economic activity. Additionally, the focus on older adults with disabilities may impact perceptions of safety and security. Therefore, age may influence feelings of safety while traveling on sidewalks. Considering the numerous factors that contribute to a person's identity is needed to better understand travel behavior and travel outcomes for public transit trips.

Mobility justice helps conceptualize the connections between diverse human needs and the risks in public spaces that affect how, why, and where people travel. This framework enables transportation planning to go beyond the study of travel outcomes to acknowledge why people do or do not travel at all. Mobility justice focuses on how identity and history shape vulnerability, which may look and feel different for each individual human being. Interview findings indicate that disabled voices and experiences have been and are currently left out of first and last mile planning. This work indicates that there is an opportunity for transportation planners to embrace a mobility justice lens to incorporate confidence, dignity, and human difference into first and last mile planning. Potential first steps include understanding the diversity of disability and considering accessibility improvements that go beyond physical improvements to make travel more efficient (wayfinding, communication, and information).

6. Conclusion

The interview findings reveal that accessible infrastructure, safe social environments, and equitable planning processes are critical to improving first and last mile travel experiences for disabled people. Despite differences in sidewalk coverage, planning priorities, and geographic location, Austin and Seattle had similar transit rider interview findings. Missing or poor-quality sidewalks, inadequate infrastructure at intersections, and car-centric street designs make disabled pedestrians experience disproportionate travel costs. While these findings offer areas for local jurisdictions to improve first and last mile conditions, it also raises tensions between how disabled people are meaningfully involved in these processes to conduct fair decision-making. From a transportation planning perspective, disabled voices are missing due to systemic failures (i.e., driver's license requirements) as well as a lack of meaningful inclusion and follow through in public engagement processes.

Interview responses suggest that perceptions of access and mobility stretch beyond what transportation planners have traditionally considered regarding accessible pedestrian and transit environments. Experiences could be improved if intersectionality and mobility justice, both of which highlight the effects of identity, history, and social structures on individual travel behavior, are better understood and acknowledged by transportation planners. Public engagement with disabled transit riders can help bridge this knowledge gap. Facilitating participatory approaches to data gathering and ensuring disabled voices are present and heard throughout the planning process can prioritize the needs of the disability community. This may include providing door-to-door transportation to and from engagement events so that existing barriers along the first and last mile do not deter individuals from participating. Focusing on meaningful and inclusive public engagement practices that empower disabled people to share their concerns can help identify locations, programs, and policies that prioritize accessibility.

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Declaration of Competing Interest

None

Appendix A. Interview Questions

- What does disability mean to you? How do you define disability?
- How do you identify yourself? This can include race, gender, sexual orientation, disability, culture, age, employment status, and income category.
- Do your identities influence your travel behavior?
- Can you think of an instance where you were treated differently when traveling because of your identities?
- What does accessibility mean to you? How do you define accessibility?
- What would an accessible neighborhood look like to you?
- What would an accessible public transit system look like to you?
- How would you design the built environment to be more accessible and inclusive?
- Which components of the sidewalk network encourage travel?
- Which components of the sidewalk network create barriers to travel?
- How do you engage with the disability community? What type of advocacy work have you participated in?
- What types of goals and investments do you think public transit agencies should prioritize?
- How frequently do you walk in your neighborhood?
- When was the last time you walked in your neighborhood? Describe that experience.
- Where do you have the best public transit experiences? What do those look like?
- Where do you experience barriers to access along your walking and/or transit trips? What are the barriers?
- For what kinds of trips do you use transit? How frequently?
- When was the last time you used transit to commute to work? Describe that experience.
- When was the last time you used transit to access healthcare? Describe that experience.
- Would you change your travel behavior if your neighborhood was built differently?
- Would you change your travel behavior if your social environment was different?
- Would you change your travel behavior if other forms of transportation were available to you? What would that look like?
- Are there any other concerns, experiences, or questions you would like to share with me?

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